

Name: Ratnesh Kumar Singh
Department: Electronics and Communication Engineering
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Phone No: +917004937129
Languages Known: English, Hindi, Bengali



Professional Summary:

Motivated Electronics and Communication Engineering student with a good foundation in programming, embedded systems, and AI. Passionate about developing innovative solutions in smart and software-driven technology.

Examination	School/College/ University	Board	Year of Completion	Specialization
10 th Standard	Vidyanjali High school (H.S.)	I.G.C.S.E	2018	Science
12 th Standard	P.B.S College	B.S.E.B	2020	Science (P.C.M.)
Bachelor in Technology	Swami Vivekananda Institute of Science & Technology	M.A.K.A.U.T. (W.B.U.T.)	2026	Electronics And Communication Engineering

Skills and Expertise:

- Programming : C,C++, PYTHON
- Web Dev: HTML, CSS
- <https://my-portfolio-omega-liart-40.vercel.app>
- UI Design: Figma
- <https://tinywebs.site/6AsQaP>

Software & Tools:

Arduino IDE, MATLAB, ArcGIS, Tinkercad, Verilog, VS Code, MS Office Suite, Canva

Special Interests:

Embedded Systems, AI, Machine Learning, Electric Vehicles, Digital Electronics

Certificates and Courses :

- Completed a one month course in Python
- Completed a one month course in C language

Training:

- Electric Vehicle Service Technician Duration- 1 month
- GIS Training Duration- 2 weeks

Projects:

- Smart Parking System: Designed a prototype using Arduino and ultrasonic sensors to detect available parking spaces efficiently.
- Designed and developed an adaptive smart antenna system to improve wireless communication reliability in high-mobility vehicular environments. Implemented beamforming algorithms that dynamically directed antenna signals toward moving vehicles, helping reduce interference and maintain stable connectivity. Combined signal processing techniques with ECE hardware components to support emerging V2X (Vehicle-to-Everything) communication standards, with applications in intelligent transportation and connected vehicle systems.
- Simple AI assistant: Created an intelligent assistant using Python and API integrations. Implemented features like command understanding, conversational responses, and real-time information retrieval.

Interests:

Exploring AI and technology trends, photography, image, audio and video editing, sketching.

Skills:

- Strong problem-solving skills
- Fast learner with excellent self-learning ability
- Highly dedicated and disciplined
- Ability to quickly adapt to new technologies

Extra Curricular Activities:

- Participated in the one day Workshop on Innovation and General Awareness in HAM Radio on 4th March 2025
- Participated in one day Industrial visit on BSNL telecom company on 24th April 2025
- Participated in the one day workshop on Empower Yourself With Cyber Awareness in Cyber Security on 8th August 2025